Indian Institute of Technology, Madras - BS in Data Science and Applications

Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

Question Paper Name :	IIT M DEGREE AN EXAM QDB2 15 Mar 2026
Subject Name :	2026 Mar15: IIT M AN EXAM QDB2
Creation Date :	2026-03-05 10:37:28
Duration :	120
Total Marks :	637
Display Marks:	Yes
Share Answer Key With Delivery Engine :	Yes
Actual Answer Key :	Yes
Calculator :	Scientific
Magnifying Glass Required? :	No
Ruler Required? :	No
Eraser Required? :	No
Scratch Pad Required? :	No
Rough Sketch/Notepad Required? :	No
Protractor Required? :	No
Show Watermark on Console? :	Yes
Highlighter :	No
Auto Save on Console?	Yes
Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Group I

Group Number :	1
Group Id :	64065332737
Group Maximum Duration :	0
Group Minimum Duration :	90

Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	637
Is this Group for Examiner? :	No
Examiner permission :	Cant View
Show Progress Bar? :	No
Revisit allowed for group Instructions? :	Yes
Maximum Instruction Time :	0
Minimum Instruction Time :	0
Group Time In :	Minutes
Revisit Section :	Yes
Action on Revisit Section :	View and Edit
Navigate To Group Summary From Last Question? :	No
Disable Submit Button During Assessment? :	No
Section Selection Time? :	0
No of Optional sections to be attempted :	0

DLP

Section Id :	640653129731
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653362924
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406531961530

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : DEEP LEARNING PRACTICE (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536373716. ✔ Yes

6406536373717. ✖ No

Sub-Section Number : 2

Sub-Section Id : 640653362925

Question Shuffling Allowed : Yes

Question Number : 2 Question Id : 6406531961531

Correct Marks : 2

Question Label : Multiple Choice Question

Consider subword tokenization methods such as Byte Pair Encoding (BPE) and WordPiece as employed in transformer-based language modeling pipelines. Which of the following statements correctly describes their fundamental properties?

Options :

6406536373718. ✖ They always split words at true morpheme boundaries (the smallest units of meaning, e.g., “un-” + “break” + “able”).

6406536373719. ✔ They learn merge rules from corpus statistics (frequency/likelihood), not token semantics

6406536373720. ✖ They eliminate (Out Of Vocabulary) OOV by ensuring every word is a single token

6406536373721. ✖ They generalize perfectly to out-of-domain text without increasing sequence length.

Question Number : 3 Question Id : 6406531961532

Correct Marks : 2

Question Label : Multiple Choice Question

Consider two subword tokenizers trained on the same text corpus. Tokenizer A uses a vocabulary of 8,000 tokens, whereas Tokenizer B uses a vocabulary of 64,000 tokens. Which of the following statements most accurately characterizes the implications of these vocabulary sizes for training efficiency in transformer-based language models?

Options :

6406536373722. ✖ Larger vocabulary always reduces sequence length and total compute.

6406536373723. ✖ Smaller vocabulary always improves semantic alignment.

6406536373724. ✔ Larger vocabulary reduces sequence length but increases embedding parameters.

6406536373725. ✖ Vocabulary size has no effect once model is pretrained.

Question Number : 4 Question Id : 6406531961533

Correct Marks : 2

Question Label : Multiple Choice Question

A subword tokenizer decomposes rare chemical entity names into a large number of fragments. During downstream fine-tuning, the model exhibits degraded performance on a chemical named entity recognition (NER) task. Which of the following represents the most principled corrective action?

Options :

6406536373726. ✖ Increase the dropout rate during fine-tuning.

6406536373727. ✔ Retrain the tokenizer using an in-domain (chemical) corpus.

6406536373728. ✖ Reduce the batch size during training.

6406536373729. ✖ Apply label smoothing to the loss function.

Question Number : 5 Question Id : 6406531961534

Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following code snippet using a pretrained BERT tokenizer:

```
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")

text = "Transformers generalize surprisingly well."

enc = tokenizer(
    text,
    max_length=12,
    padding="max_length",
    truncation=True,
    return_tensors="pt"
)

print(len(enc["input_ids"][0]))
print(tokenizer.convert_ids_to_tokens(enc["input_ids"][0]))
```

Which of the following statements must necessarily be true about the token sequence produced by this tokenizer configuration?

Options :

6406536373730. ✖ The output sequence will always contain exactly 12 non-padding tokens.

6406536373731. ✔ Special tokens may consume a portion of the 12-token length budget.

6406536373732. ✖ Word boundaries are preserved, since truncation occurs at whitespace boundaries.

6406536373733. ✖ Padding will never be applied when truncation is enabled.

Question Number : 6 Question Id : 6406531961535

Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following code snippet using a pretrained BERT tokenizer:

```
from transformers import AutoModelForMaskedLM

model = AutoModelForMaskedLM.from_pretrained("bert-base-uncased")

outputs = model(
    input_ids=batch_ids,
    attention_mask=batch_mask,
    labels=batch_ids
)

loss = outputs.loss
loss.backward()
optimizer.step()
```

Assume this loop is run on a large domain corpus without labeled task targets. What is the most precise interpretation?

Options :

6406536373734. ✖ This is supervised fine-tuning with noisy labels.

6406536373735. ✖ This is instruction tuning because labels are present.

6406536373736. ✔ This performs continual pretraining with MLM objective.

6406536373737. ✖ This trains a classifier head implicitly.

Question Number : 7 Question Id : 6406531961536

Correct Marks : 2

Question Label : Multiple Choice Question

In the standard Transformer architecture (e.g., GPT or BERT), which component exhibits a computational complexity that scales quadratically ($O(N^2)$) with respect to the input sequence length N?

Options :

6406536373738. ✖ The initial token and positional embedding lookup.

6406536373739. ✖ The element-wise addition in the Residual (Skip) connections.

6406536373740. ✔ The computation of the attention score matrix (

QK^T).

6406536373741. ✖ The linear projections in the Position-wise Feed-Forward Network (FFN).

Question Number : 8 Question Id : 6406531961537

Correct Marks : 2

Question Label : Multiple Choice Question

A model with 4B parameters (fp32) is being trained using the AdamW optimizer on a single 32GB GPU. Why is full fine-tuning impossible in this configuration?

Options :

6406536373742. ✖ The 4B parameters alone require 32GB of VRAM, leaving no room for the OS or CUDA kernels.

6406536373743. ✔ The optimizer states (32GB) plus the model weights (16GB) and gradients (16GB) exceed the 32GB VRAM limit.

6406536373744. ✖ The tokenizer is unable to map a context length of 2048 to a 32-bit integer space.

6406536373745. ✖ The fp32 precision requires 8 bytes per parameter, meaning the 4B model needs 64GB just to load.

Question Number : 9 Question Id : 6406531961538

Correct Marks : 2

Question Label : Multiple Choice Question

During the full fine-tuning of a Large Language Model, you aim to regularize the training process by penalizing the growth of weight magnitudes. This ensures the model does not overfit the fine-tuning data by making excessively large updates to the pretrained parameters. Which parameter is specifically designed to apply this penalty?

Options :

6406536373746. ✖ Gradient Accumulation.

6406536373747. ✖ Batch Size.

6406536373748. ✖ Checkpointing.

6406536373749. ✔ Weight Decay.

Question Number : 10 Question Id : 6406531961539

Correct Marks : 2

Question Label : Multiple Choice Question

Why does full fine-tuning of large language models often require more memory than inference using the same model?

Options :

6406536373750. ✖ Inference uses larger batch sizes.

6406536373751. ✖ Inference stores optimizer states.

6406536373752. ✔ Training requires storing gradients and optimizer states.

6406536373753. ✖ Training uses longer input sequences.

Question Number : 11 Question Id : 6406531961540

Correct Marks : 2

Question Label : Multiple Choice Question

Which technique is most effective at reducing GPU memory usage during training without modifying the model architecture?

Options :

6406536373754. ✖ Gradient clipping.

6406536373755. ✔ Mixed-precision training.

6406536373756. ✖ Increasing learning rate warmup.

6406536373757. ✖ Saving checkpoints less frequently.

Sub-Section Number : 3

Sub-Section Id : 640653362926

Question Shuffling Allowed : Yes

Question Number : 12 Question Id : 6406531961541

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the output of a Hugging Face tokenizer when applied to a batch of sequences with padding= "max_length" and return_tensors= "pt". Which of the following statements are correct? (Select ALL that apply)

Options :

6406536373758. ✔ The input_ids tensor contains the numerical indices mapped from the vocabulary.

6406536373759. ✔ The attention_mask contains 0s for padding tokens and 1s for real tokens to prevent the model from attending to padding.

6406536373760. ✖ The token_type_ids (if present) are used primarily to distinguish between uppercase and lowercase letters.

6406536373761. ✔ In models like BERT, the input_ids will typically begin with a special token index (e.g., [CLS]).

Question Number : 13 Question Id : 6406531961542

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the following training configuration for fine-tuning a large language model:

```
TrainingArguments(
    per_device_train_batch_size=1,
    gradient_accumulation_steps=8,
    fp16=True,
    max_grad_norm=1.0
)
```

Which of the following effects are expected from this configuration? (Select ALL that apply)

Options :

6406536373762. ✔ Effective batch size is larger than 1.

6406536373763. ✔ Reduced memory usage compared to FP32 training.

6406536373764. ✔ Gradient clipping limits large parameter updates.

6406536373765. ✖ Gradient accumulation reduces the number of optimizer states.

Question Number : 14 Question Id : 6406531961543

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements correctly distinguish different Transformer architectures and their typical training objectives? (Select ALL that apply)

Options :

6406536373766. ✔ Encoder-only models are commonly trained with Masked Language Modeling objectives.

6406536373767. ✔ Decoder-only models rely on causal masking and autoregressive inference.

6406536373768. ✔ Encoder-decoder models are well suited for sequence-to-sequence tasks.

6406536373769. ✖ All Transformer architectures use identical attention masks.

Question Number : 15 Question Id : 6406531961544

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the following code used to fine-tune a decoder-only (causal) language model:

```
def format_example(example):
    return {
        "text": (
            "Instruction: Classify sentiment\n"
            "Input: " + example["sentence"] + "\n"
            "Answer: " + example["label_text"]
        )
    }

dataset = raw_dataset.map(format_example)

trainer = Trainer(
    model=model,
    train_dataset=dataset,
)
```

Compared to using a standard AutoModelForSequenceClassification (which adds a linear layer to the final hidden state to predict a fixed set of label IDs), how does the generative setup change the training behavior? (Select ALL that apply)

Options :

6406536373770. ✔ Task Conditioning: The model learns to follow natural language formatting (Instruction/Input/Answer) rather than just mapping a sequence to a single integer ID.

6406536373771. ✔ Open Vocabulary: The model's prediction head remains the size of its full vocabulary (e.g., 50k+ tokens), allowing it to generate any string as an answer instead of being restricted to fixed logits for N classes.

6406536373772. ✖ Guaranteed Zero-Shot: This training format ensures the model will generalize perfectly to any unseen task prompt without further data.

6406536373773. ✔ Objective Shift: The training objective moves from minimizing cross-entropy loss over a discrete class index to minimizing next-token prediction loss over the sequence tokens.

Question Number : 16 Question Id : 6406531961545

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

A model is trained using the Causal Language Modeling (CLM) objective with a standard cross-entropy loss. Which of the following statements correctly describe the training dynamics? (Select ALL that apply)

Options :

6406536373774. ✔ The loss at position t depends only on tokens $< t$.

6406536373775. ✔ The attention mask is strictly upper triangular.

6406536373776. ✖ Future tokens contribute gradients to earlier positions.

6406536373777. ✔ The joint probability of the sequence is factorized autoregressively.

Question Number : 17 Question Id : 6406531961546

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

When configuring TrainingArguments for a Transformer model, we include a learning rate warmup phase (e.g., warmup_steps=500). Which of the following statements correctly describe the purpose and behavior of this strategy? (Select ALL that apply)

Options :

6406536373778. ✔ It involves linearly increasing the learning rate from 0 (or a small value) to the target maximum during the initial phase.

6406536373779. ✔ It helps prevent "divergence'' or instability caused by large gradients when the model weights are far from their optimal state.

6406536373780. ✖ It significantly reduces the VRAM (memory) required to store optimizer states.

6406536373781. ✔ After the warmup phase, the learning rate typically follows a decay schedule (like linear or cosine) to ensure convergence.

Sub-Section Number :	4
Sub-Section Id :	640653362927
Question Shuffling Allowed :	Yes

Question Number : 18 Question Id : 6406531961547

Correct Marks : 4

Question Label : Short Answer Question

A transformer-based language model has 1.2 billion parameters and is fine-tuned using AdamW. Assume parameters, gradients, and optimizer states (first and second moments) are all stored in 32-bit precision (4 bytes each). Ignoring activations and buffers, calculate the total GPU memory required (in GB) to store parameters, gradients, and optimizer states. (Assume 1 GB = 10^9 bytes.)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

19.1 to 19.3

Question Number : 19 Question Id : 6406531961548

Correct Marks : 4

Question Label : Short Answer Question

A GPT-style transformer block has an embedding dimension of 768 and uses 12 attention heads.

Consider a specific processing task with a very short sequence of only 3 tokens ($T = 3$).Compute the total number of weight parameters (ignore biases) in the self-attention module, specifically for the Query, Key, Value, and Output projection matrices. Report your answer in millions, rounded to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

2.3 to 2.5

Question Number : 20 Question Id : 6406531961549

Correct Marks : 4

Question Label : Short Answer Question

You are training a decoder-only language model with a vocabulary size of 32,000, maximum context length of 2048, and embedding dimension of 1024. The model uses learned positional embeddings. Calculate the total number of parameters in the embedding layer (token + positional embeddings). Report your answer in millions, rounded to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

34.7 to 35.1